

EE3005 Communication Systems - Assignment 5

Department of Electrical Engineering, IIT Madras

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Topics covered: Pulse code modulation, delta modulation, M-ary digital modulation, PSK, QAM, ML detection, probability of error

Instructions: Submit your solutions in one PDF file with all step-by-step answers, plots and outputs written/printed in that file.

1. A PCM system uses a uniform quantizer followed by an 8-bit encoder. The system operates with a bit rate of 96×10^6 bps.
 - a) Determine the maximum message bandwidth for which the system operates without aliasing.
 - b) If the input signal is a full-scale sinusoidal wave, compute the SQNR in dB.
 - c) If the amplitude range of the input signal is ± 4 V, calculate Quantization step size and Maximum quantization error
2. A speech signal of duration 15 seconds is sampled at a rate of 10 kHz and encoded using a uniform PCM system. The required SQNR is 48 dB.
 - a) Determine the minimum number of bits per sample required.
 - b) Calculate the total number of bits needed to store the signal.
3. A BPSK system transmits symbols $s = \pm\sqrt{E_b}$ with $E_b = 2$. The received sample after matched filtering is:

$$r = s + n$$

where $n \sim \mathcal{N}(0, 0.5)$.

1. What symbol is detected using ML rule?
 2. Compute the probability that this decision is incorrect given $s = +\sqrt{E_b}$ was transmitted.
4. A BPSK system transmits $s = \pm\sqrt{E_b}$ with $E_b = 4$. The received signal is:

$$r = s + n, \quad n \sim \mathcal{N}(0, 1)$$

Due to receiver bias, the decision threshold is $\gamma = 0.5$.

1. Write the decision rule.
2. Compute the probability of error.
3. Compare with the optimal detector.

5. A QPSK system transmits:

$$I, Q \in \{\pm 1\}$$

Noise variances:

$$\sigma_I^2 = 1, \quad \sigma_Q^2 = 4$$

Received:

$$r_I = 0.2, \quad r_Q = -0.5$$

1. Detect the transmitted symbol.
2. Compute probability of error in each branch.
3. Identify the dominant error source.

6. An 8-PSK system operates at:

$$\frac{E_b}{N_0} = 10 \text{ dB}$$

1. Convert to E_s/N_0 .
2. Compute symbol error probability:

$$P_s \approx 2Q \left(\sqrt{\frac{2E_s}{N_0}} \sin \frac{\pi}{8} \right)$$

3. Estimate bit error probability.

7. Consider the octal signal-point constellations shown in the figure

- a) The nearest neighbor signal points in the 8-QAM signal constellation are separated in distance by A units. Determine the radii a and b of the inner and outer circles.
- b) The adjacent signal points in the 8-PSK are separated by a distance of A units. Determine the radius r of the circle.
- c) Determine the average transmitter powers for the two signal constellations and compare the two powers. What is the relative power advantage of one constellation over the other? (Assume that all signal points are equally probable).
- d) Assign three data bits to each point of the signal constellation such that nearest (adjacent) points differ in only one bit position.
- e) Determine the symbol rate if the desired bit rate is 90 Mbps.
- f) Compare the SNR required for the 8-point QAM modulation with that of an 8-point PSK modulation having the same error probability.

- g) Which signal constellation, 8-point QAM or 8-point PSK, is more immune to phase errors? Explain the reason for your answer.

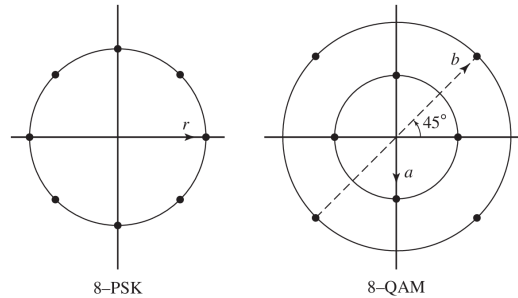


Figure 1: Octal signal-point constellations

8. Suppose that binary PSK is used for transmitting information over an AWGN with $N_0 = 10^{-10} \text{ W/Hz}$. The transmitted signal energy is $E_b = \frac{A^2 T}{2}$, where T is the bit interval and A is the signal amplitude. Determine the signal amplitude required to achieve an error probability of 10^{-6} , if the data rate is (a) 10 kbps, (b) 100 kbs, (c) 1 Mbps.
9. Consider a digital communication system that transmits information via QAM over a voice-band telephone channel at a rate 2400 symbols/second. The additive noise is assumed to be white and Gaussian. Determine the E_b/N_0 required to achieve an error probability of 10^{-5} at: (a) 4800 bps, (b) 9600 bps, (c) 19200 bps.
10. Consider the four-phase and eight-phase signal constellations shown in Figure. Determine the radii r_1 and r_2 of the circles such that the distance between two adjacent points in the two constellations is d . From this result, determine the additional transmitted power required in the 8-PSK signal to achieve the same error probability as the four-phase signal at high SNR, where the probability of error is determined by errors in selecting adjacent points.

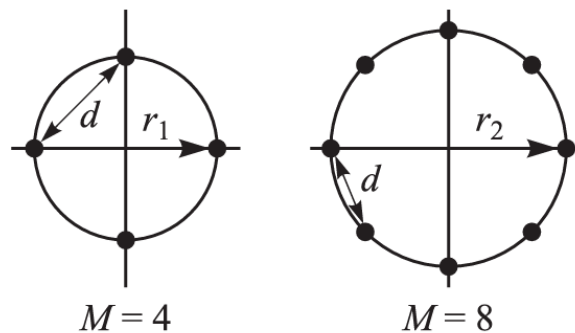


Figure 2: four-phase and eight-phase signal constellations

11. The 16-QAM signal constellation shown in Figure is an international standard for telephone-line modems (called V.29). Determine the optimum decision boundaries for the detector, assuming that the SNR is sufficiently high so that errors only occur between adjacent points.

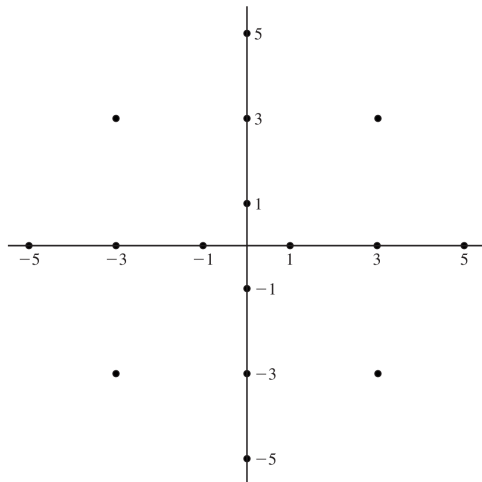


Figure 3: 16-QAM signal constellation (V.29)

12. Assume that the set of lowpass signals

$$\{\phi_{nl}(t), n = 1, 2, \dots, N\}$$

forms an orthonormal basis for representing the lowpass-equivalent signals

$$\{s_{ml}(t), m = 1, 2, \dots, M\}.$$

Define

$$\phi_n(t) = \sqrt{2} \Re \{ \phi_{nl}(t) e^{j2\pi f_0 t} \}$$

and

$$\tilde{\phi}_n(t) = -\sqrt{2} \Im \{ \phi_{nl}(t) e^{j2\pi f_0 t} \}.$$

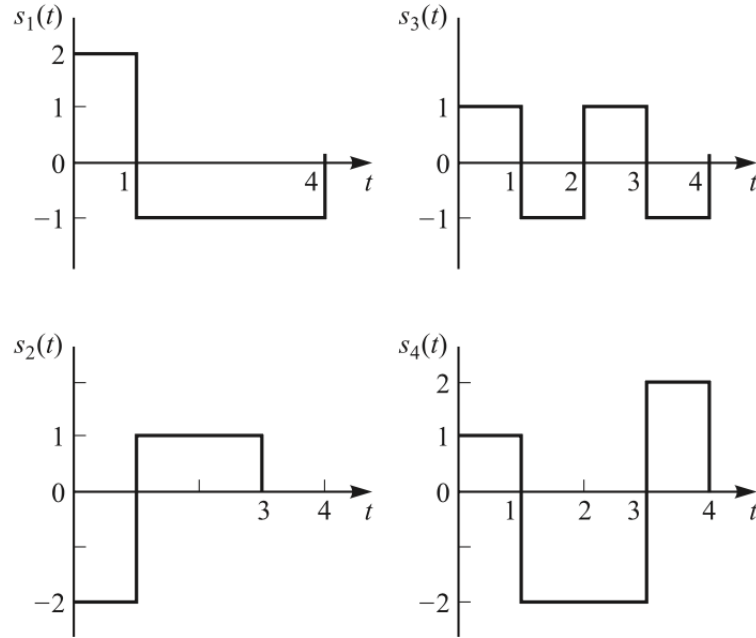
Show that the set

$$\{\phi_n(t), \tilde{\phi}_n(t), n = 1, 2, \dots, N\}$$

forms a $2N$ -dimensional orthonormal basis sufficient to represent the bandpass signals

$$s_m(t) = \Re \{ s_{ml}(t) e^{j2\pi f_0 t} \}, \quad m = 1, 2, \dots, M.$$

13. Consider the four waveforms shown in Figure P2.11.



1. Determine the dimensionality of the waveforms and a set of basis functions.
 2. Use the basis functions to represent the four waveforms by vectors s_1 , s_2 , s_3 , and s_4 .
 3. Determine the minimum distance between any pair of vectors.
14. Two equiprobable messages are transmitted over an additive white Gaussian noise channel with

$$\frac{N_0}{2} = 1.$$

The two signals are

$$s_1(t) = \begin{cases} 1, & 0 \leq t \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$s_2(t) = s_1(t - 1).$$

The receiver is implemented using a correlation-type structure. However, due to an imperfection in the upper branch, the receiver computes

$$r_1 = \int_0^{1.5} r(t) dt, \quad r_2 = \int_1^2 r(t) dt.$$

Determine the optimal decision rule based on the observations r_1 and r_2 .

15. A communication system employs binary antipodal signals

$$s_1(t) = \begin{cases} 1, & 0 < t < 1, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$s_2(t) = -s_1(t).$$

The received signal is

$$r(t) = s(t) + As(t - 1) + n(t),$$

where $s(t)$ is the transmitted signal, A is an exponential random variable with density

$$f_A(a) = \begin{cases} e^{-a}, & a > 0, \\ 0, & \text{otherwise,} \end{cases}$$

and $n(t)$ is additive white Gaussian noise with power spectral density $N_0/2$. The delay of the multipath component is equal to 1, and A and $n(t)$ are independent. The two messages are equiprobable.

1. Determine the optimal decision rule.
2. Compare the error probability with that of a system without multipath.

16. Three 8-ary signal constellations are shown in the below Figure.

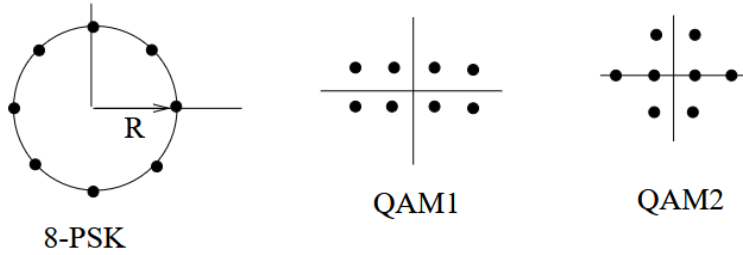


Figure 5: Signal constellations for

- (a) Express R and $d_{\min}^{(2)}$ in terms of $d_{\min}^{(1)}$ so that all three constellations have the same E_b .
 - (b) For a given E_b/N_0 , which constellation do you expect to have the smallest bit error probability over a high SNR AWGN channel?
 - (c) For each constellation, determine whether you can label signal points using 3 bits so that the label for nearest neighbors differs by at most one bit. If so, find such a labeling. If not, say why not and find some "good" labeling.
17. Consider an M -ary digital communication system where $M = 2^N$, and N is the dimension of the signal space. Suppose that the M signal vectors lie on the vertices of a hypercube that is centered at the origin, as illustrated below. Determine the average probability of a symbol error as a function of E_s/N_0 , where E_s is the energy per symbol, $N_0/2$ is the power spectral density of the AWGN, and all signal points are equally probable.

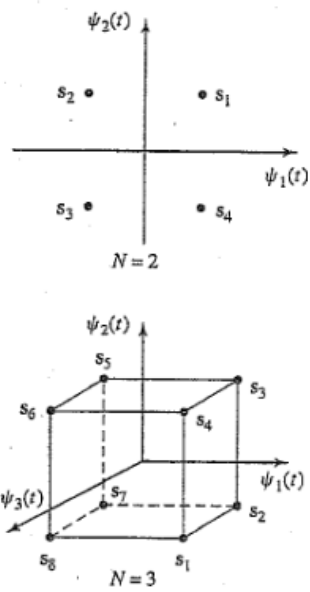


Figure 6: Signal constellations